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An Efficient High Performance GDI based 4-bit Vedic Multiplier in 32nm Technology

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***Abstract*—System performance improves with good multiplier performance. In today's digital world, the speed of the multiplier is a crucial factor, but it also has a high power consumption. Power and delay optimisation of multipliers will be pivotal. The choice of adder units in the multiplier also takes into account adders such the Ripple carry adder(RCA), carry look-ahead adder(CLA), and carry skip adder(CSA). This research investigates the differences between several adders implemented in Vedic multiplier using CMOS, transmission gate and GDI based techniques. When attempting to design a Vedic multiplier with CMOS transistors and Transmission gates, the circuit will experience problems such as delay, power and area metrics. These problems can be addressed by employing the Gate Diffusion Input (GDI) logic. This uses GDI logic to keep delay, power, and the number of transistors as low as feasible. In this research, a GDI-based 4-bit Vedic multiplier was built in 32nm technology and compared to a CMOS and TG-based vedic multiplier design. GDI-based 4-bit Vedic Multiplier systems were faster, used less power, and took up less area than CMOS and TG-based systems. This 4-bit Vedic multiplier design can be used in the Tanner EDA tool with different adders.**

***Index Terms*—CMOS, Transmission Gate(TG), Gate Diffusion Input (GDI), Ripple carry adder(RCA), carry look-ahead adder(CLA), and carry skip adder(CSA), Delay and Power.**

I. Introduction

Multipliers are crucial in modern digital signal processing and related fields. The addition and multiplication of two binary values are the most basic and frequently utilised arithmetic operations in high performance systems. The majority of DSP algorithms and over 70% of microprocessor instructions involve arithmetic operations. Thus, the processing times are not a problem. Because of this, an efficient multiplier is essential[1]. Growing technology and signal processing needs have increased the importance of fast processing. An additional design consideration for multipliers is low power consumption. Because dynamic power accounts for a disproportionate share of total power consumption, the demand for high-speed, low-power multipliers has grown in recent years[2]. The primary focus of a designer is on creating power-efficient, high-speed circuits. The goal of designing an excellent multiplier is to create a compact, fast, and low-power device[3].

Vedic mathematics refers to a body of mathematical theory that originated in ancient India and was rediscovered in the early twentieth century. The sixteen principles or word-formulae that form the basis of Vedic mathematics are referred to as Sutras. Based on the effectiveness of the Vedic multiplication algorithm in digital multipliers, authors speculate on the potential utility of Vedic mathematics in digital signal processing[4]. In this article, the Urdhva Triyakbhyam (Vertically and Crosswise) Sutra can be used to inform the design of a simple forming digital multiplier, or Vedic multiplier. This Sutra was frequently used in ancient India to multiply two decimal numbers. This Sutra is introduced, and then it is applied to the binary number system such that it may be employed in digital hardware[5]. We discuss the hardware of a vedic multiplier and show how it is related to the more common array multiplier. Many more applications of technology may also become apparent [6] if the ancient Veda text is explored in detail.

A Vedic multiplier using RCA (Ripple Carry Adder), CLA (Carry Look-Ahead Adder), and CSA (Carry Select Adder) combines these different adder architectures to perform multiplication efficiently. This kind of multiplier can be used to multiply two binary numbers using various adder structures to optimize speed and performance. The primary advantage of this approach is that it combines the strengths of RCA, CLA, and CSA adders. The RCA is simple to implement but slow, while CLA and CSA adders are faster but more complex. By using CLAs or CSAs for the partial products and the final addition, you can achieve both speed and efficiency in the Vedic multiplier. A 4-bit GDI-based Vedic multiplier is a type of digital circuit that performs multiplication using the ancient Vedic multiplication technique, which is an ancient system of Indian mathematics. Gate Diffusion Input (GDI) is a popular

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low-power digital logic style. In this context, we are creating a 4-bit multiplier circuit using GDI logic that leverages Vedic mathematics principles.

II. RELATED WORKS

Both the CMOS-based methods using a variety of adders and the Transmission gate-based Vedic multiplier presented by Swati T. Mestry et al. [7] were recognised in significant technologies. By reducing the operating voltage by 20%, the TG-based CLA adder sees reductions of 76.33 percent in average power, 48.13 percent in power delay product, and 13.69 percent in energy-delay product. When moving from 25 degrees Celsius to 50 degrees Celsius, the TG-based CLA adder's average power drops to 41.42 percent, while the power delay product rises to 65.33 percent and the energy delay product falls to 93.28 percent. In order to improve the efficiency of the multiplier, Aruru Sai Kumar et al., [8] demonstrated a novel 8-bit vedic multiplier that employs Parallel Prefix adders, most widely known as the Brent Kung adder. In place of alternative methods used for multiplication, the Urdhva Tiryagbhyam sutra is now being used because it is applicable to all cases of algorithms for N x N bit integers and offers the lowest latency. This justifies its current use. An 8-bit vedic multiplier can be built out of one OR gate, two 8-bit Brent Kung adders, one 4-bit Brent Kung adder, and four 4-bit vedic multipliers. However, by combining the aforementioned elements, this is possible. When compared to previous attempts with similar features, the speed provided by the newly built Vedic Multiplier is far more effective.

There is a 4-bit Vedic multiplier that runs on GDI logic that was made by Ankit Garg et al., [9]. Two-bit Vedic multipliers built in CMOS and GDI are compared in terms of their transient reaction to see how well the GDI logic works. The purpose of this comparison is to find out how well the GDI reasoning works. Two different designs are now being looked at as possible alternatives to the 4-bit Vedic multiplier. When we judge how well these circuits work, we look at how much power they use on average, how much delay they have, and how much temperature they can handle. More study is being done to see if it is possible to change the voltage coming from the supply. We use the bulk metal oxide semiconductor field effect transistor predictive technology model to look at the device characteristics. The circuit computations are done at a precision of 130 nm. P. Saritha et al., [10] produced the concepts for a 4-bit Vedic multiplier on 18nm FinFET, and they utilised the various design methodologies that we discussed. A GDI-based AND, half adder, and full adder circuit can be utilised to build a 4-bit Vedic multiplier if the first way, which is referred to as Design 1. This must be done in order to bring the job to a successful conclusion. It is intended that a 4-bit Vedic multiplier (Design 2) will be created with the assistance of a 2-bit Vedic multiplier that is based on GDI, a half adder, and a 4-bit Ripple carry circuit. This will be done. Because of this, it will be possible to construct the multiplier. These take into consideration typical values for things like average power, usual latency, typical transistor density, and typical board size.

Two new symmetric designs for low-power full adder cells with a GDI (Gate-Diffusion Input) structure and a hybrid CMOS logic style were developed by Vahid Foroutan et al., [11]. These patterns are mirror images of one another. These adder modules were designed with complete voltage swing support as a main need, alongside low power dissipation and high speed. An initial design has been created using a mixed logic strategy. To create new complete adders with the required performance, the hybrid logic style integrates aspects of multiple different logic types. This provides the designer with more leeway in their design, letting them optimise for a wider variety of use cases with less work. The second concept is an innovative take on the standard adder cell layout that eliminates the need for XOR and XNOR gates. In addition to the complete voltage swing circuit, an Ultra Low-Power, high-speed digital component is provided by including the GDI (Gate-Diffusion-Input) approach into the design.

III. EXISTED 4-BIT VEDIC MULTIPLIER

The uniqueness of Vedic mathematics is that it eliminates the lengthy and very simple equations using Urdhva Triyagbhyam (Vertically and Crosswise) Sutra one of 16 Vedic mathematical sutras. The Vedic sutra, known as Urdhva Triyagbhyam means both vertically and crosswise. This sutra is a powerful way to multiply and results in a faster multiplication by partial generation of the product and a summing up in a single iterative step[12-16]. This sutra has the advantage of saving considerable time and effort to solve the problem.

In the traditional method of multiplication, the partial products are not summarised until after each individual partial product has been obtained in a horizontal fashion. In the Vedic method of multiplying, partial products are summed vertically while more sums are added simultaneously until all of the items in the column have been collected. Fig. 1 is a depiction of the block diagram for the 4-bit vedic multiplier.

A. CMOS based 4-bit Vedic Multiplier

Digital circuit that multiplies two 4-bit binary values according to the rules of Vedic mathematics and is implemented in CMOS technology[17-20]. CMOS technology is commonly used in modern digital integrated circuits due to its low power consumption and high performance. It consists of input registers to hold the two 4-bit operands, a series of shift-and-add stages, and an output register to store the result. In each shift-and-add stage, CMOS logic gates and multiplexers are used to perform bit-wise additions, shifts, and carry generation as per the "Urdhva Tiryagbhyam" Vedic multiplication technique. The final result is obtained after multiple stages of processing, making it an efficient and parallel method for 4-bit multiplication in CMOS technology. The Full Adder design using CMOS technology is depicted in Fig. 2.

B. TG based 4-bit Vedic Multiplier

A 4-bit Vedic multiplier implemented using transmission gate technology utilizes transmission gates (also known as pass

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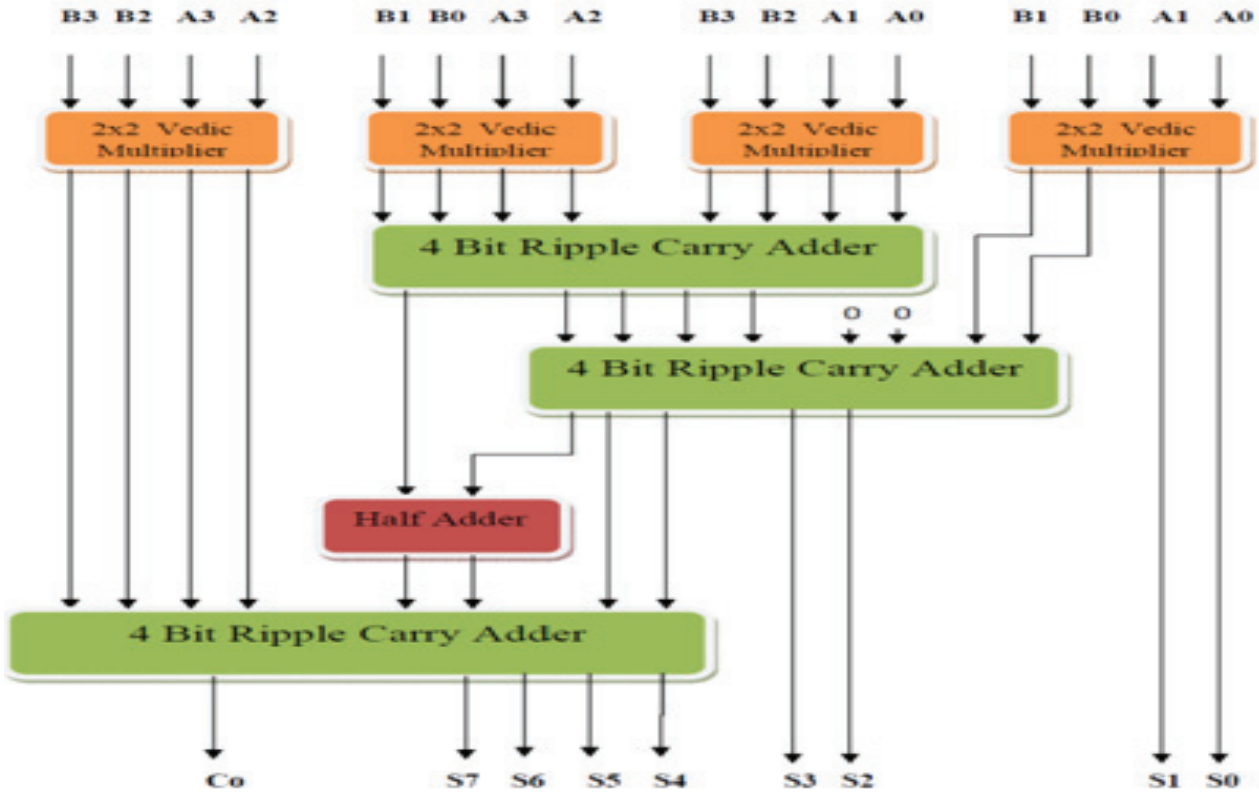


Fig. 1. Block diagram of 4-bit vedic multiplier using RCA

TABLE I
COMPARISION OF DELAY, POWER AND TC FOR CMOS, TG AND GDI BASED 4-BIT VEDIC MULTIPLIER

Parameter	CMOS			Transmission Gate(TG)			Proposed Gate Diffusion Input(GDI)		
	RCA	CLA	CSA	RCA	CLA	CSA	RCA	CLA	CSA
Delay(ns)	45.8	66.3	47.07	41.7	63.8	46.7	37.5	55.6	32.3
Avg Power(μW)	35.6	105.6	37.3	26.5	87.9	29.6	24.8	78.5	27.3
Transistor Count(TC)	828	1032	900	444	1104	516	206	386	278

gates) for performing the multiplication operations. Transmission gates are complementary CMOS-based switches used to pass or block signals[21-23]. In this design, each bit of the two 4-bit numbers is processed using a series of transmission gates. The inputs pass through these gates, and the logical operations such as AND, XOR, and OR can be implemented using these gates to achieve the crosswise and vertical multiplication steps specified in Vedic mathematics. The transmission gates allow for efficient and low-power logic operation, and the design can be optimized for speed and area utilization[24-25]. However, a full description of a 4-bit Vedic multiplier using transmission gates would involve detailed schematic diagrams, which can be quite complex, but this approach can be efficient for low-power applications. The Full Adder design using TG technology is depicted in Fig. 3.

IV. PROPOSED 4-BIT VEDIC MULTIPLIER USING GDI TECHNOLOGY

Gate Diffusion Input (GDI) is a low-power design method that reduces static power consumption while simultaneously enhancing logic swing. This method allows for the implementation of several logic functions with a reduced number of transistors. Since fewer transistors are required in this technique (in comparison to TG and CMOS), it is well-suited to the design of quick, low-power circuits.This logic style may be manufactured using a conventional CMOS process, and it improves upon prior technology in terms of output voltage, power, and power delay product. Figure 4 depicts the GDI implementation of the Full Adder design.

Vedic multiplication and GDI can both be seen to be effective in distinct ways. A smaller number of transistors can be employed in a circuit's design using GDI logic, while the time it takes to complete an operation can be cut in half with

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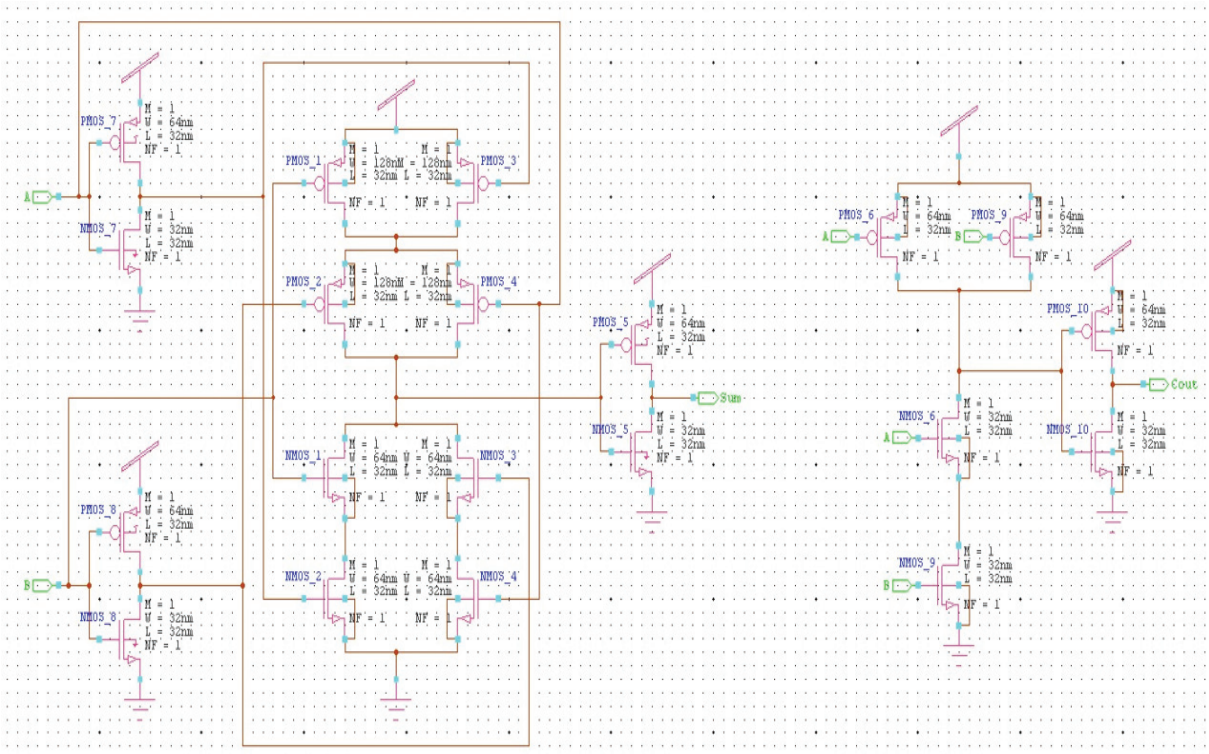


Fig. 2. Full Adder design using CMOS technology

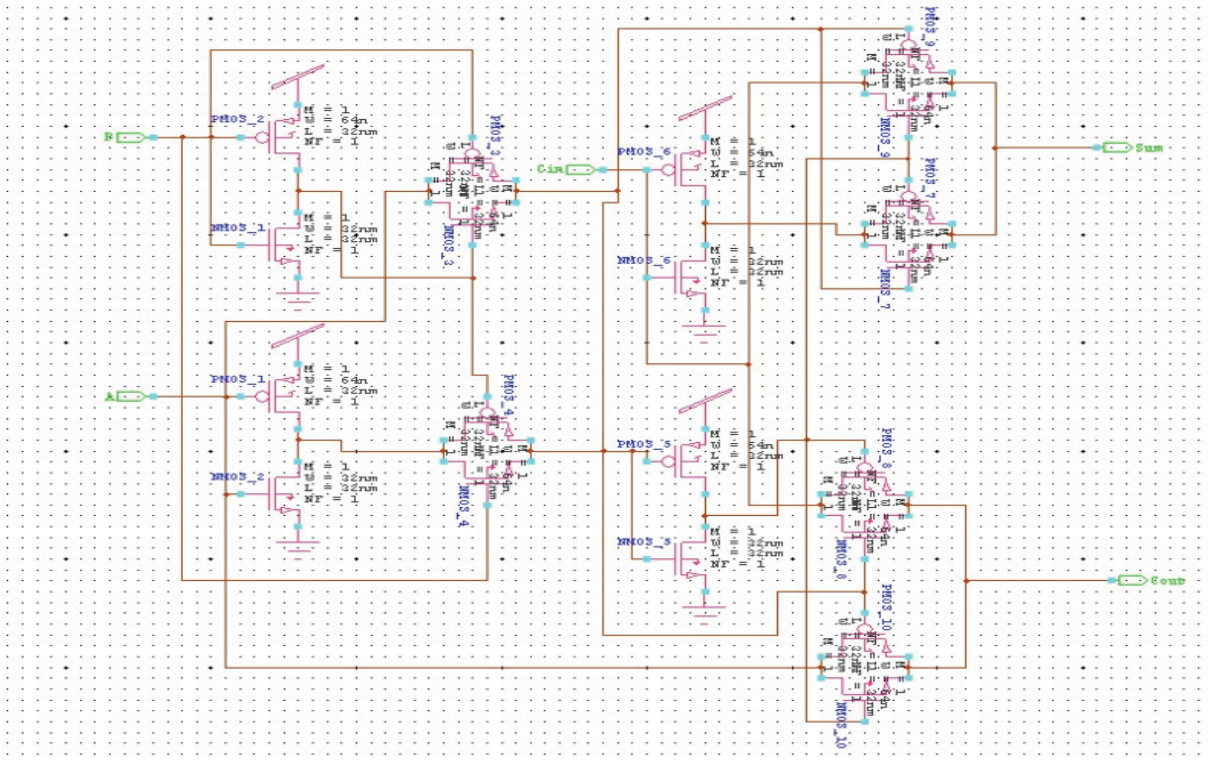


Fig. 3. Full Adder design using Transmission Gate(TG) technology

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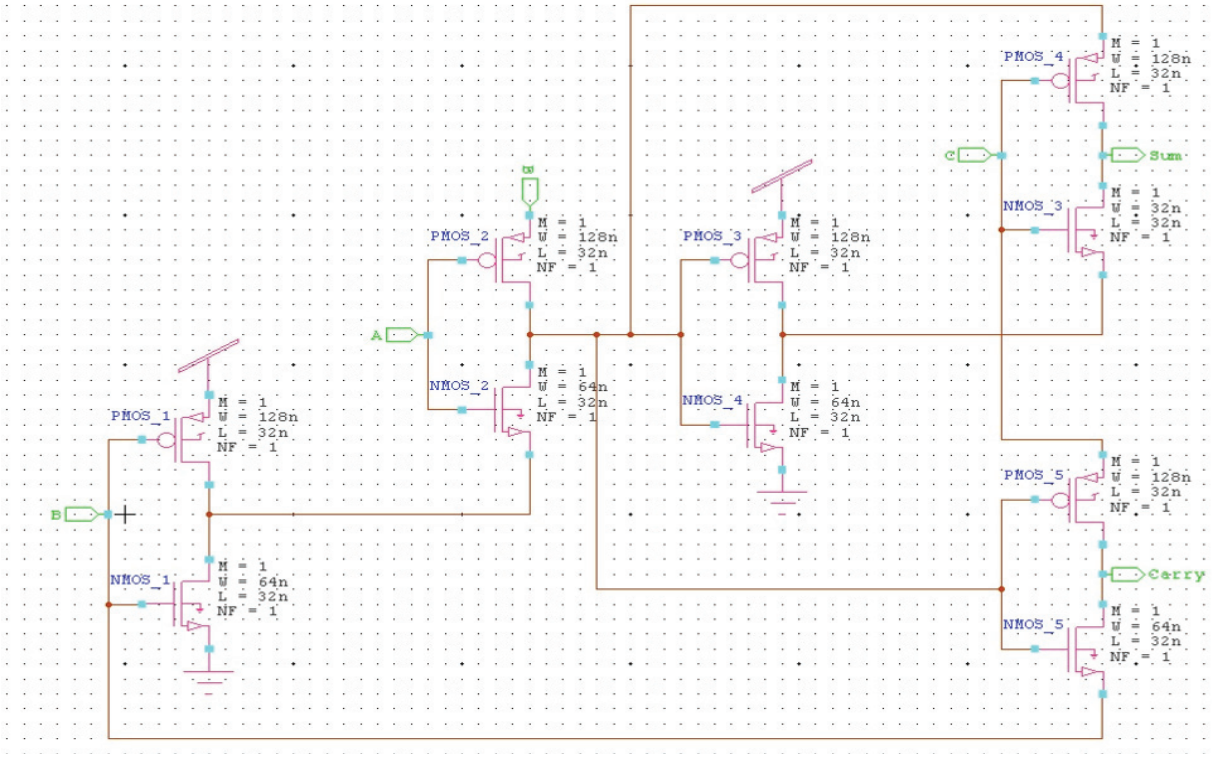


Fig. 4. Full Adder design using Gate Diffusion Input (GDI) technology

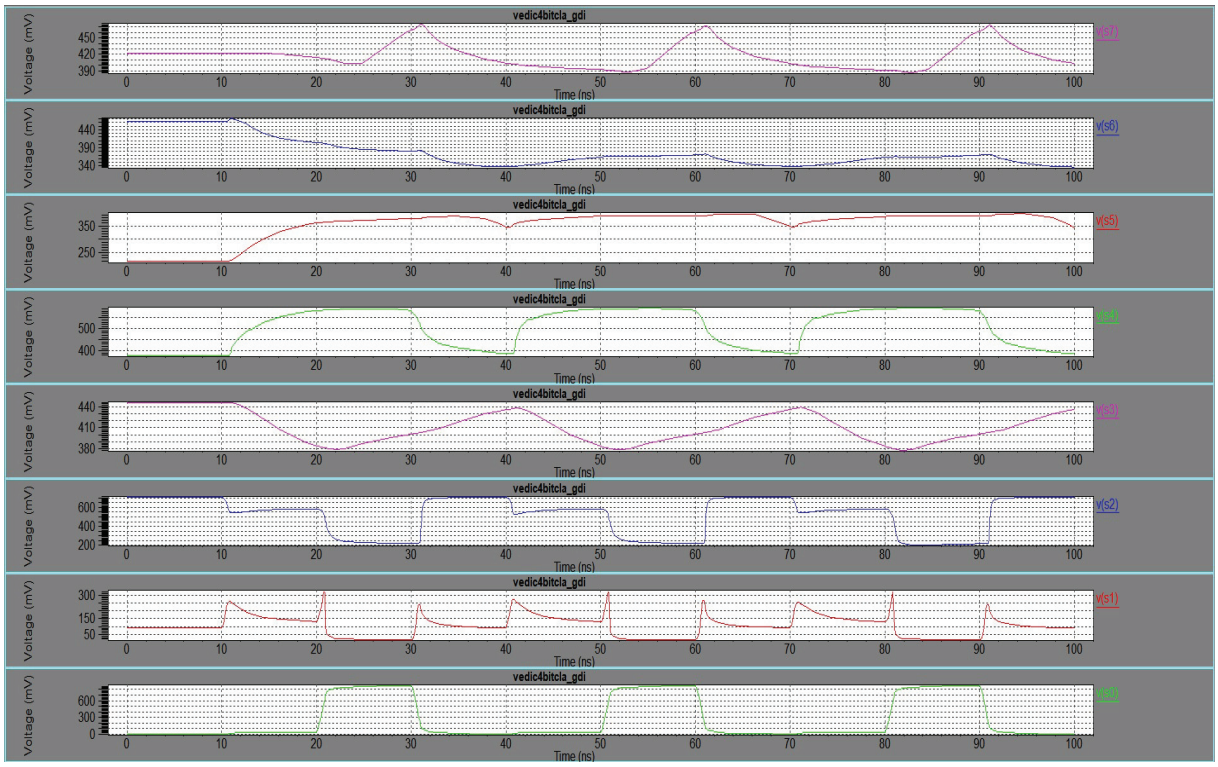


Fig. 5. Simulation waveform of 4-bit vedic multiplier using GDI Technology

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the help of vedic multiplication. Since there will be fewer transistors, the switching node capacitance and the amount of logical effort needed to switch will both drop, leading to a lower dynamic power consumption. This means there is cause for optimism for the GDI-based Vedic multiplier to be effective.

V. Simulation Results

The 32nm Gate Diffusion Input (GDI) 4-bit Vedic Multiplier was invented. CMOS-based and TG-based vedic multipliers were compared using several adders. This research discusses Ripple carry adders (RCA), Carry lookahead adders (CLA), and Carry skip adders (CSA). Tanner EDA was used to create the multiplier design with several adders. Figure 5 shows a GDI-simulated 4-bit vedic multiplier waveform. Table 1 compares the performance parameters of the proposed GDI-based 4-bit Vedic Multiplier, CMOS, and TG multipliers. Latency, average power, and area are measured in nanoseconds, microwatts, and transistor counts, respectively.

VI. Conclusion and Future work

In this research, the differences between various adders (RCA, CLA, and CSA) that were developed for Vedic Multiplier utilising CMOS, transmission gate, and GDI approaches are investigated and analysed. The GDI-based 4-bit Vedic multiplier that was proposed was built in 32nm technology, and it was compared with CMOS and TG-based vedic multiplier designs that utilised a variety of adders.The proposed GDI-based 4-bit Vedic Multiplier outperformed CMOS and TG circuits in delay, power consumption, and area.

Later, researchers can examine the GDI-based Vedic multiplier's suitability for designing complex signal processing circuits such convolutors for convolution algorithms and floating point multipliers. The multiplier's Vedic characteristics would make the final outcome more accurate, enabling this occur.

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